Deep Learning for Cross-Sectional Stock Ranking

A sequence-to-rank pipeline with quant-aware evaluation

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Final Project Presentation

noisy sequence modeling → learning-to-rank → portfolio construction. [1].

[1] https://github.com/runqi-allen-wang/Deep-Sequential-Models-for-Cross-Sectional-Stock-Selection

1. Problem framing: not price prediction, but sequence-to-rank learning

Machine-learning view

For stock i at date t, the input is a multivariate time-series window

$$X_{i,t} \in \mathbb{R}^{L \times d}, \qquad d = 47,$$

and the neural network outputs one scalar score

$$s_{i,t} = f_\theta(X_{i,t}).$$

The score is used only for **ranking** stocks on the same date.

Why this is hard for DL

- weak and noisy labels;
- temporal distribution shift;
- high-dimensional panel data;
- non-differentiable downstream decision: top-k selection.

Core idea: convert a financial task into a supervised learning-to-rank problem with realistic out-of-sample evaluation.

2. Supervision: rank labels reduce non-stationary scale effects

Cross-sectional rank target

Let $r_{i,t+h}$ be the future h-day return. The label is

$$y_{i,t} = 2\,\mathrm{rankpct}_{j\in\mathcal{A}}(r_{j,t+h}) - 1 \in [-1,1].$$

This converts raw return regression into a normalized daily ranking task.

Signal diagnostic

$$\mathrm{IC}_t = \mathrm{Corr}_{i\in\mathcal{A}}(s_{i,t}, y_{i,t}).$$

IC is the cross-sectional correlation between model scores and future ranks.

ML interpretation

- Rank labels act like date-wise normalization.
- The model learns relative strength, not absolute price levels.
- IC measures global ordering quality, analogous to a differentiable rank proxy.

3. Quant layer: from alpha score to active return

Alpha-score interpretation

The neural score is treated as a cross-sectional alpha signal:

$$\hat{\alpha}_{i,t} = \mathrm{zscore}_{i \in \mathcal{A}}(s_{i,t}).$$

It is not a price forecast; it is a relative signal used to rank assets.

Portfolio map

$$w_t = \Pi_{\mathcal{C}}\big(\phi(\hat{\alpha}_t)\big),$$

where ϕ is top-k selection and $\Pi_{\mathcal{C}}$ encodes long-only / weighting constraints.

Quant-aware utility

For benchmark weights b_t,

$$R^{\mathrm{act,net}}_{t+1} = w_t^\top r_{t+1} - b_t^\top r_{t+1} - \frac{c}{10000}\mathrm{Turn}_t.$$

What this adds to ML

- IC evaluates signal quality.
- Active return evaluates benchmark-relative utility.
- Turnover penalizes unstable decisions.

4. Objective mismatch: differentiable training vs. top-k evaluation

Training is a surrogate problem

We optimize differentiable losses such as regression, correlation, and pairwise ranking losses:

$$\min_{\theta} \ \mathbb{E}_{(i,t)} \, \ell(f_\theta(X_{i,t}), y_{i,t}).$$

Evaluation is a decision problem

The final portfolio is a thresholded decision rule:

$$S_t = \mathrm{TopK}_{i \in \mathcal{A}}(s_{i,t}), \qquad w_{i,t} = \frac{\mathbf{1}\{i \in S_t\}}{|S_t|}.$$

Main technical difficulty

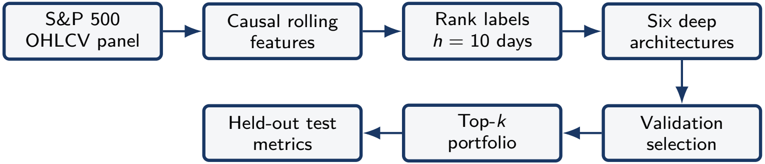
A model can improve average ranking correlation, but still fail if the improvement does not occur in the extreme upper tail selected by TopK.

Transaction-cost layer

$$\mathrm{Turn}_t = \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|,$$

$$R_t^{\mathrm{net}} = R_t^{\mathrm{gross}} - \frac{c}{10000} \mathrm{Turn}_t.$$

5. Pipeline: a reproducible ML experimental protocol



Computer-science controls

- chronological train/validation/test split;
- no future leakage in rolling features;
- model-wise hyperparameter search;
- three random seeds.

Quant evaluation metrics

- IC / Rank signal: cross-sectional alpha quality;
- net and excess return: benchmark-relative utility;
- turnover: trading frequency and cost exposure;
- long-short spread: top-vs-bottom separation.

6. Model zoo as inductive-bias comparison

Model	Mathematical form	What it tests
DLinear	$s = w^\top \mathrm{vec}(X) + b$	Is a linear signal already enough?
ResMLP	$h^{\ell+1} = h^\ell + g_\ell(h^\ell)$	Are nonlinear feature interactions useful?
GRU/LSTM	$h_\ell = \mathrm{Gate}(x_\ell, h_{\ell-1})$	Does learned memory help under noisy temporal patterns?
TCN	$y_\ell = \sum_{m=0}^{K-1} W_m x_{\ell-dm}$	Are local temporal motifs captured by dilated convolution?
PatchTST	$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d_k})V$	Can patch attention model long-window interactions efficiently?
Ensemble	$s = \frac{1}{M}\sum_m \mathrm{zscore}(s^{(m)})$	Does model averaging improve stable global ranking?

Design principle

The experiment is not only “which model wins”, but which **inductive bias** best matches low-SNR financial sequences.

7. Why PatchTST-style tokenization is a natural fit

Patch construction

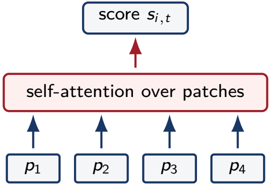
Split the length-L window into patches $p_k \in \mathbb{R}^{P \times d}$, then project

$$u_k = W_p \mathrm{vec}(p_k) + b_p.$$

The sequence length changes from L to approximately L/P tokens.

Attention complexity

Day-level attention: $O(L^2 d_k)$. Patch-level attention: $O((L/P)^2 d_k)$.



Intuition for non-experts

A patch is a short local pattern, like recent momentum or volatility. Attention then compares such patterns across the lookback window.

8. Loss design: robust regression + ranking regularization

Robust base loss

$$\ell_\delta(s, y) = \begin{cases} \frac{1}{2}(s-y)^2/\delta, & |s-y| \le \delta, \\ |s-y| - \frac{1}{2}\delta, & |s-y| > \delta. \end{cases}$$

SmoothL1 is less sensitive to outlier labels than MSE.

IC-aware loss

$$\mathcal{L}_{\mathrm{IC}} = 1 - \mathrm{Corr}_{i \in \mathcal{B}_t}(s_{i,t}, y_{i,t}).$$

Pairwise ranking loss

$$\mathcal{L}_{\mathrm{pair}} = \mathbb{E}_{(i,j)} \log(1 + \exp(-(s_i - s_j)\mathrm{sign}(y_i - y_j))) .$$

Combined objective

$$\mathcal{L} = \mathcal{L}_{\mathrm{SmoothL1}} + 0.1\mathcal{L}_{\mathrm{IC}} + 0.05\mathcal{L}_{\mathrm{pair}}.$$

This is a differentiable approximation to a non-differentiable ranking-and-selection pipeline.

9. Quant-aware model selection: validation is multi-objective

Why validation cannot use loss alone

A lower supervised loss may still produce a poor strategy if it causes unstable ranks or weak top-tail separation. Therefore model selection uses a validation objective that combines signal and decision metrics.

Schematic validation score

$$\mathrm{ValScore} = z(R^{\mathrm{net}}) + z(R^{\mathrm{excess}}) + z(\mathrm{IC}) + z(R^{\mathrm{LS}}) - \lambda z(\mathrm{Turn}).$$

Quant interpretation

This is a small, reproducible version of an alpha-research loop:

1. learn a signal;
2. convert it into holdings;
3. evaluate after costs;
4. select by validation performance, not test results.

The ML model is judged by the utility of the decision rule it induces.

10. Main empirical result: PatchTST wins the decision-level metric

Model	Test IC	Long-only return	Excess return	Turnover
DLinear	0.0234	9.83%	−7.42%	0.2218
Ensemble	**0.0512**	18.25%	1.00%	0.2087
GRU	0.0270	21.51%	4.25%	**0.1057**
LSTM	0.0272	19.66%	2.40%	0.1201
PatchTST	0.0496	**25.42%**	**8.17%**	0.1945
ResMLP	0.0395	17.22%	−0.04%	0.3452
TCN	0.0311	22.71%	5.46%	0.2364

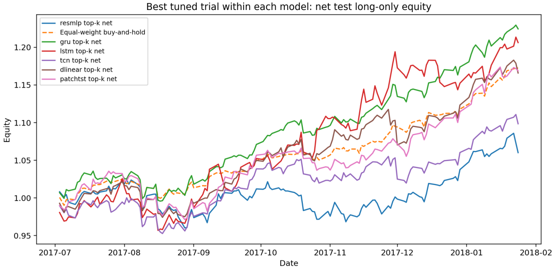
Result

PatchTST achieves 25.42% $\pm$ 4.68% net long-only return and 8.17% $\pm$ 4.68% excess return over buy-and-hold.

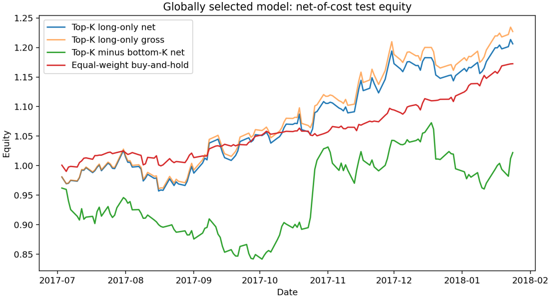
Important nuance

The ensemble has the highest IC, but lower top-k return. Better global correlation is not sufficient for better top-tail decisions.

11. Visual check: equity curves, not only table values



Model-family winners, seed 0

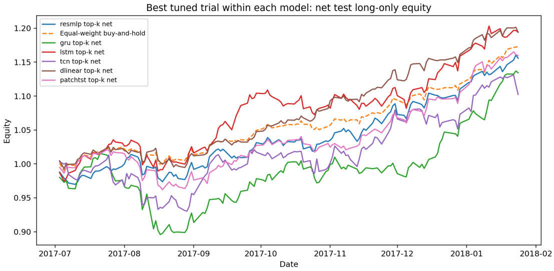


Global validation winner, seed 0

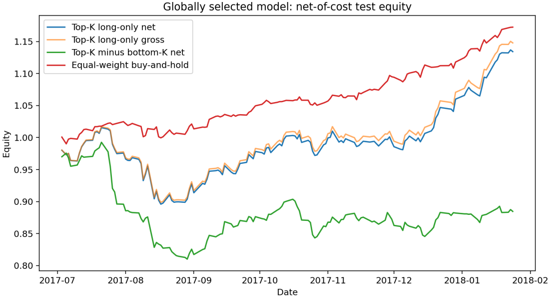
What to look for

The equity curve checks whether performance is accumulated over time or caused by one short episode. The gross–net gap visualizes the transaction-cost penalty.

12. Visual check: equity curves, not only table values



Model-family winners, seed 1

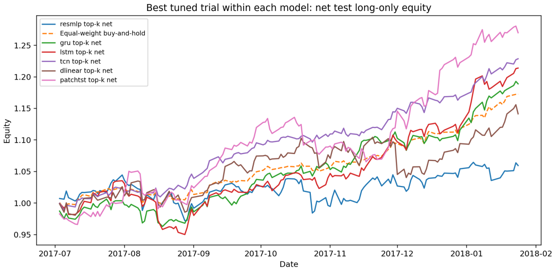


Global validation winner, seed 1

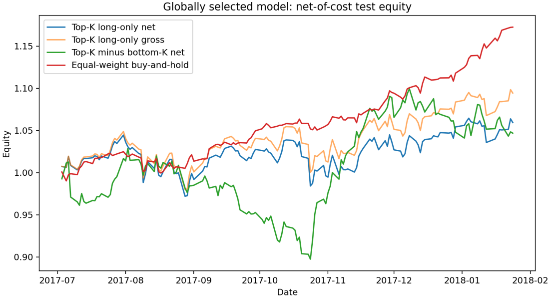
What to look for

The equity curve checks whether performance is accumulated over time or caused by one short episode. The gross–net gap visualizes the transaction-cost penalty.

13. Visual check: equity curves, not only table values



Model-family winners, seed 2



Global validation winner, seed 2

What to look for

The equity curve checks whether performance is accumulated over time or caused by one short episode.
The gross–net gap visualizes the transaction-cost penalty.

14. Ablation I: global ranking improvement does not guarantee utility

Loss	Test IC	Long-only return	Excess return	Turnover
SmoothL1	0.0117	**19.22%**	**1.97%**	**0.1289**
SmoothL1+IC	0.0149	17.44%	0.18%	0.2216
SmoothL1+Pairwise	**0.0326**	18.46%	1.20%	0.3616
SmoothL1+IC+Pairwise	0.0203	15.57%	−1.68%	0.1800

ML lesson

Pairwise ranking improves IC from 0.0117 to 0.0326, so the surrogate objective does improve global ordering.

Decision lesson

It also increases turnover and does not give the best net top-k return. Surrogate alignment is incomplete.

15. Ablation II: post-processing is part of the algorithm

Portfolio rule	Long-only return	Excess return	Turnover
Equal, no smoothing	17.44%	0.18%	0.2216
Equal, EMA(0.7)	**18.89%**	**1.64%**	**0.1541**
Inv-vol, no smoothing	17.85%	0.60%	0.2681
Inv-vol, EMA(0.7)	18.83%	1.57%	0.1959

Score smoothing as algorithmic stability

$$\bar{s}_{i,t} = \alpha s_{i,t} + (1-\alpha)\bar{s}_{i,t-1}.$$

Smoother scores reduce rank switching, selected-set instability, and turnover.

CS/ML takeaway

The deployment rule is not a detail after prediction; it is an algorithmic layer that changes generalization at the decision level.

16. Takeaways, limitations, and what is novel here

Core contributions

1. A clean sequence-to-rank formulation for stock selection.
2. An inductive-bias comparison across linear, MLP, recurrent, convolutional, Transformer, and ensemble models.
3. Quant-aware evaluation connecting IC, active return, long-short spread, turnover, and transaction costs.

Limitations

- public S&P 500 data may contain survivorship bias;
- only daily OHLCV features are used;
- this is an empirical course project, not a live trading system.

Final message

The key deep-learning question is not whether a network can fit noisy returns, but whether its learned representation survives ranking, top-k selection, temporal shift, and transaction costs.

References

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